

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fifth Semester of B. Tech. Examination (C.E.)

November 2012

CE-302 System Programming (S.P.)

Date: 28.11.2012, Wednesday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Rough work is to be done in the last page of main supplementary, please don't write anything on the question paper.
5. Indicate clearly, the option(s) you attempt along with its respective question no.
6. Figures to the right indicate marks.

SECTION-I

Q-1 Answer the following questions.

1. Write a note on the execution of a program, explaining all the steps. 3
2. Is a compiler a translator? Are the linker and loader a translator? Justify. 3
3. Is there any interface between users and computer hardware (system)? Is there any interface between computer hardware (system) and system software? Explain in brief. 4

Q-2

- [A] Are application software and system software dependent on machine structure? Give examples of application software and system software. What are the differences between system programming and system software? 4
- [B] What are the components of system software? What are the various system programs that are commonly used in modern PCs? Give system programming hierarchy. 4
- [C] What is a preprocessor? How does it differ from macro-processor? Differentiate I brief between preprocessor and macro-processor with all the aspects. 4

OR

Q-2

- [A] What does the preprocessor instruction #include in 'C' do? Does the macro-processor do the same kinds of tasks? Why? 4
- [B] Why do we study the following parameters? 4
1.Main Memory 2.Registers 3.Data Formats 4.Instruction Format
- [C] Describe the different functions of loaders. What do you mean by linkage (linking) loader? How to design an absolute loader? When is a linker necessary? Explain. 4

Q-3

- [A] What are the attributes and contents of the symbol table? What is the data structure used to implement a symbol table in an efficient way? 3
- [B] What is an assembler? What are the functions of assembler? List the various assembler directives. What is OPTAB? What is SYMTAB? What is location counter? 5
- [C] Why is it necessary to relocate a program in the memory? Define loading, relocating and linking. Define relocating loaders. 4

OR

Q-3

- [A] What is a debugger? How debugger helps in the development process? 4
- [B] What is Editor? What are various functions performed by the Editor? What are the various types of Editors? Why text editors are required? 4
- [C] Differentiate between system call and system interrupts. Differentiate between open macros and closed macro. 4

SECTION-II

Q-4

- 1. What are the differences between a file and a library? 3
- 2. What are the advantages and disadvantages of holding symbolic operation codes in a separate symbol table? 4
- 3. Prove that a two-pass assembler is always feasible. 4

Q-5

- [A] What do you mean by program relocation? What is known as linking loader? 3
- [B] What is a macro? Define macro instruction, expanding of macros and macro invocation. Is macro different from a subprogram? How? In what context does a macro have to be used? 5
- [C] Programming languages are usually compiled into the machine code of the target computer, but sometimes an interpretive system is used. Discuss the relative merits and demerits of these two systems. 4

OR

Q-5

- [A] Give reasons, why compiler is typically broken down into phases and passes. Indicate a typical set of data structures passed between them. 4
- [B] Differentiate between: Compiler, Interpreter, Token and Lexeme, LEX and YACC, Compile Time and Run. 4
- [C] If the expression $A=B*C + D*E$ is input to the compiler, what would be the functions performed by each and every phase. What will be the outcome after each and every phase? 4

Q-6

- [A] When implementing a compiler, why is a regular language commonly used for lexical analysis phase and why context free grammars are used for the main syntax analysis phase? 4
- [B] What are the major functions done by a Compiler? What are the major challenges in the design of the modern compiler? Explain machine independent and machine dependent phases of compiler. 4

OR

- [B] How does a programmer decide whether to use a macro or a subroutine to accomplish a given logical functions? What are the similarities and differences between the use of macros at translation time and the use of subroutine at execution time? 4
- [C] State the responsibilities of the linker in program development? Why is linking sometimes deferred until execution time? What will happen when you perform static linking with shared libraries? 4

OR

- [C] Explain linking process with a block diagram. Under what circumstances the combined tasks of the linker and the loader are useful? 4